

# Intro and problem definition



*10-701 Introduction to Machine Learning*  
*Geoff Gordon*

# ***Logistics***

- <https://10-701.github.io>
  - ▶ This whole website is required reading
- Highlights:
  - ▶ Tech: Piazza, Gradescope, Canvas
  - ▶ Grades: homeworks (written and coding parts), quizzes (during recitation slot, check understanding of lectures and HWs), midterm & final exams
  - ▶ Prereqs: probability, coding, derivatives, reasoning/proofs
  - ▶ Grace days for homeworks: no need to ask us about small illnesses, trips, etc.
    - applied greedily, limit on total lateness
    - separate from major problems (e.g., longer-lasting illness)

# ***Collaboration***

- Is encouraged and can support learning!
- But write up solutions ***on your own***
  - ▶ don't discuss full concrete solutions, just strategies and ideas
  - ▶ maybe best not to take detailed notes during discussions, but if you have notes, put them away, wait 10 minutes before writing anything
  - ▶ ensures solution isn't coming from short-term memory
- Make sure to ***protect your own work***
- And ***disclose all collaborators and sources***
- We do check, and we regularly find problems — see syllabus about academic integrity violations



# *GenAI*

- You are encouraged to use GenAI to improve learning!
  - ▶ e.g., find literature, generate practice problems, discuss strategy — let us know if you find an interesting and helpful study practice!
- All work that you submit must be your own
- The idea: do it at least once yourself, so that you know what the hard parts are and what can go wrong
  - ▶ then have GPT or Claude do stuff for you after the end of 10-701, and you'll be able to catch subtle problems

# *GenAI*

- Treat GenAI as if it were another student:
  - ▶ interactions should support learning
  - ▶ do not discuss complete solutions
  - ▶ best not to take written notes during discussion
  - ▶ all notes should be closed while you are writing solutions
- This includes turning off large-scale code completion: you will have trouble with quizzes if you yourself don't learn what the libraries do (and also with understanding and debugging code after the end of the course)

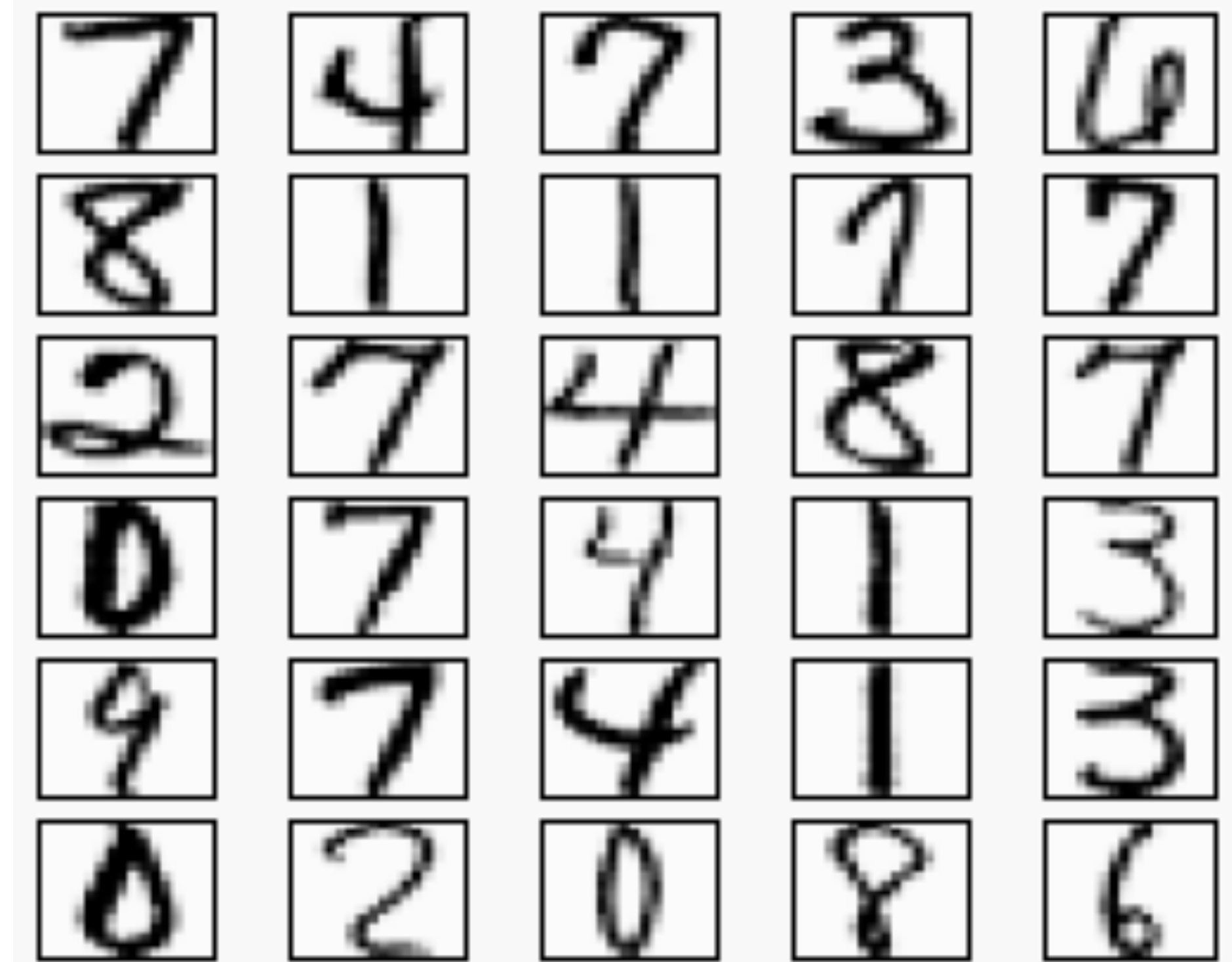
```
scipy.sparse.linalg.eigs(A, k=6, M=None,  
    sigma=None, which='LM', v0=None,  
    ncv=None, maxiter=None, tol=0,  
    return_eigenvectors=True, Minv=None,  
    OPinv=None, OPpart=None, rng=None)
```

# ***What is ML?***

- Code that gets better at some ***task*** by taking advantage of some kind of ***experience***
  - ▶ Task is formalized with a performance ***metric***
  - ▶ Experience is formalized with ***data***



*For example*



- Task: route mail by reading handwritten ZIP code
- Experience: mail that has been correctly routed
- Becomes:
  - ▶ data: images of individual digits w/ corresponding labels
  - ▶ metric: per-digit accuracy (%)

## ***Another example***

- Learning whether to approve a loan or line of credit
  - ▶ Task: *predict good loan*
  - ▶ Experience: *previous loan decisions*
  - ▶ Data: *income, other debts*
  - ▶ Metric: *accuracy*



# ***No single answer***

- Often the same/similar ML problem can be formulated several ways
  - ▶ previous slide's metric: accuracy of approval decision (classification)
  - ▶ could also predict a numerical credit score (regression)
  - ▶ or distribution of total amount repaid (conditional density estimation)
  - ▶ or each loan payment over time (sequence learning)
  - ▶ ...

*No single answer*

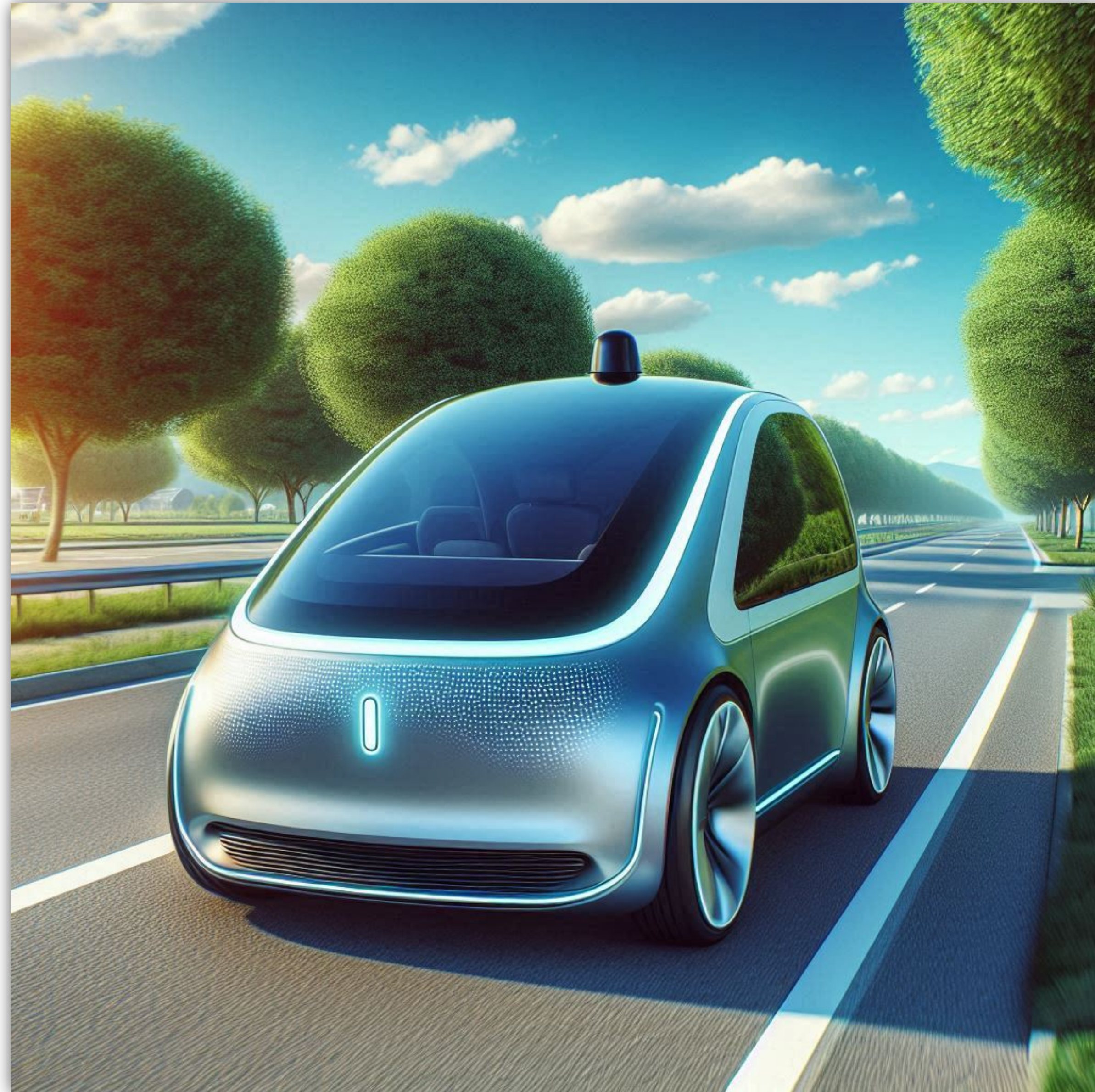
- Often several
- ▶ previous (classification)
- ▶ could be multiple related
- ▶ or discrete estimation
- ▶ or each
- ▶ ...

| Output type        | Name                                           |
|--------------------|------------------------------------------------|
| boolean            | Binary classification                          |
| categorical        | Multiclass classification                      |
| real               | Regression                                     |
| ordering           | Ranking                                        |
| multiple related   | Structured prediction (e.g., graphical models) |
| multiple over time | Sequence learning                              |
| policy / control   | Reinforcement learning                         |
| ...                | ...                                            |

related  
classification  
regression)  
density  
ing)



# ***Self-driving car***





***What is our  
experience?  
How do we  
turn it into  
data?***



road time - video, features  
steering angle  
speed  
lidar

local laws

weight

time of day

weather



# ***What is our task? How do we measure performance?***



# times polled over  
time to destination  
miles to intervention  
# accidents  
learned supervisors - lane departure  
accuracy of state estimation  
loss of traction  
imitation accuracy  
\$1  
☆☆☆☆

***Your turn:  
well-posed  
ML  
problems***

**Task**

medical diagnosis  
galaxy classification

**Experience**

MD experience  
images from surveys  
(crowdsourced labels)

**Performance metric**

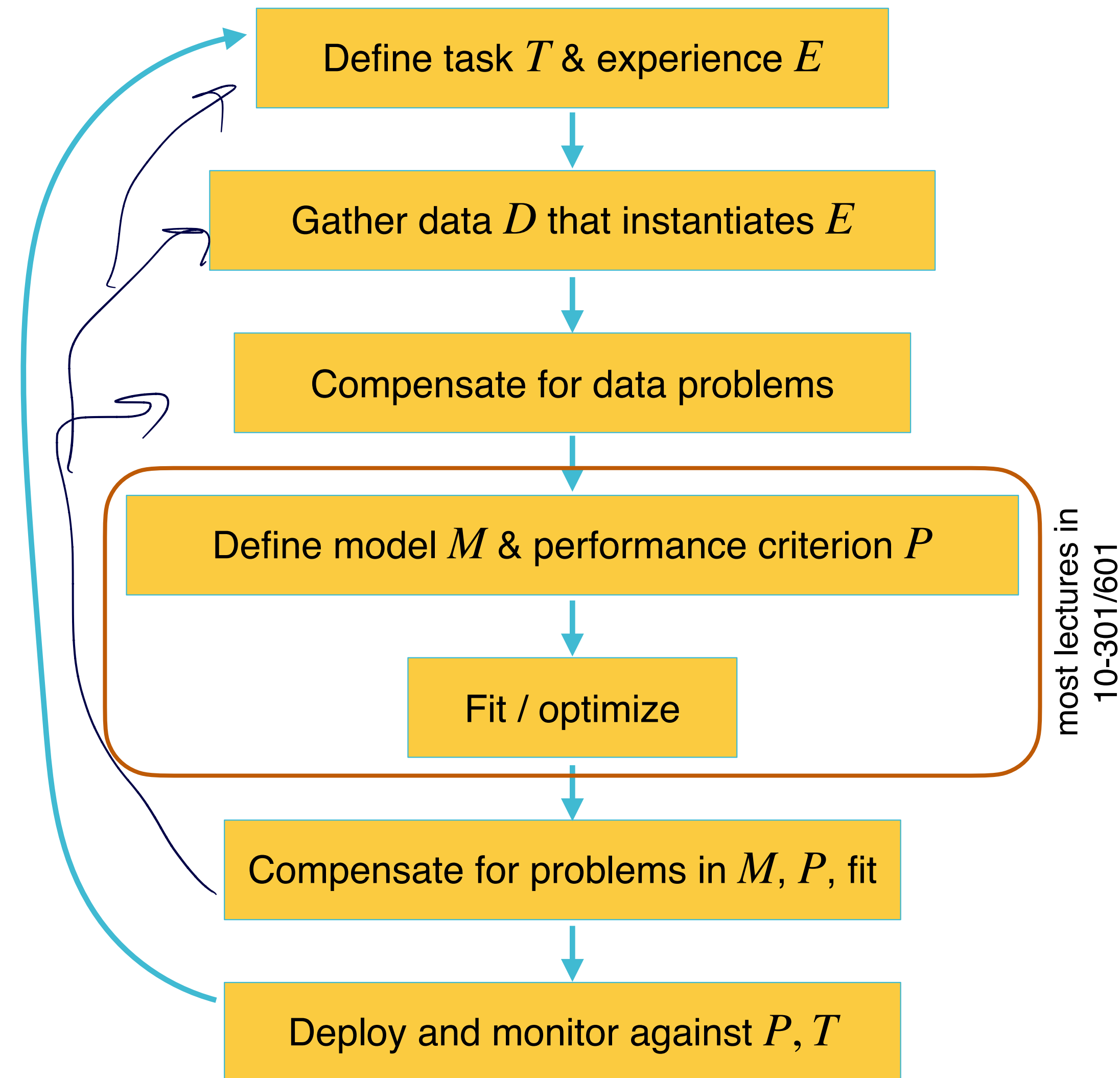
accuracy  
accuracy/distortion

**Data**

patient records  
pixels, (consensus labels)



# ML pipeline



*what do we want to accomplish?*

*define input and output representations*

*distribution shift, admixture (outliers), mislabeling, ...*

*how do we represent our hypothesis? how do we measure success?*

*hyperparameters, regularization, ...*

*model mismatch, distribution shift, overfitting, bad local optima, safety, laws, business rules ...*

*we never get it right the first time — learner does what we tell it, not what we should have told it!*

# ***A brief history of machine learning***

- Early Foundations (1940s–1960s)
  - ▶ 1957: Rosenblatt develops the perceptron, an early neural network for classification
- Symbolic AI & the First AI Winter (1970s–1980s)
  - ▶ Limitations of perceptrons (Minsky & Papert, 1969) and lack of computing power lead to skepticism and reduced funding
- Statistical & Algorithmic Advances (1980s–1990s)
  - ▶ 1986: Rumelhart, Hinton & Williams popularize backpropagation, enabling multi-layer neural networks to learn
  - ▶ 1980s–90s: Emergence of support vector machines (SVMs), decision trees, boosting (AdaBoost), Bayesian methods, reinforcement learning

# ***A brief history of machine learning***

- Second AI Winter, Rise of Data (1990s–2000s)
  - ▶ Disillusionment with difficulty of training deep networks and problems with scaling RL, eventually overcome by...
  - ▶ Explosion of digital data + faster computing power
  - ▶ Kernel methods, ensemble methods come into their own
- Deep Learning Revolution (2010s)
  - ▶ 2012: AlexNet (Krizhevsky, Sutskever, Hinton) wins ImageNet competition w/ GPU + deep CNNs, igniting deep learning boom
  - ▶ Reinforcement learning breakthroughs (e.g., AlphaGo in 2016)
- Foundation Models & Generative AI (2020s–present)
  - ▶ Rise of transformers (Vaswani et al., 2017) revolutionizes NLP (BERT, GPT)
  - ▶ Emergence of foundation models trained on massive datasets for general-purpose use
  - ▶ Policy, ethics, and responsible AI practices gain prominence due to societal impacts



# ***Classical and modern models***

- Huge amount of change over time in the types of problems and types of models that people focus on
- But, some ideas and techniques span the whole history
  - ▶ how to set up problems, useful mathematical ideas, relationship to optimization, even just overall ML mindset
- One of our goals this course: expose these common ideas so that you can develop new models & techniques and be part of whatever the next revolution is
  - ▶ simple example: linear models vs. last layer of deep net

# ***Techniques***

- Some of the common techniques:
  - ▶ linear algebra
  - ▶ differential calculus
  - ▶ optimization (e.g., SGD)
  - ▶ hypothesis classes, complexity, regularization
  - ▶ tensor manipulation
  - ▶ function spaces
  - ▶ ...

***Driving  
(some of  
the)  
differences:  
how much  
data,  
compute***

- Data:
  - ▶ deep nets can get very high performance with big data, can lose badly to classical methods like feature engineering + logistic regression if we have small data
  - ▶ kernel methods are intermediate (may need more data than simple classical models, but don't seem to be SOTA on huge data)
  - ▶ small data still happens
    - e.g., if expensive to collect
    - e.g., long tail



***Driving  
(some of  
the)  
differences:  
how much  
data,  
compute***

- Compute:
  - ▶ kernel methods can be prohibitively expensive on big data (there are approximations that can trade off accuracy)
  - ▶ deep nets are driving a huge boom in construction of data centers (and corresponding electricity use)
  - ▶ small compute still happens (e.g., phones, sensors)
- Extreme compute limitation: models that humans can run in their heads (e.g., MMSE for evaluating cognitive impairment)
  - ▶ for interpretability, trust, unplugged use

*Our first ML task*

- Learning to diagnose heart disease ( $T$ )
  - ▶ as a **(supervised) binary classification task** ( $E$ )

|             |                |                        |             |                |       |
|-------------|----------------|------------------------|-------------|----------------|-------|
|             | features       |                        |             | labels         |       |
|             | Family History | Resting Blood Pressure | Cholesterol | Heart Disease? |       |
| data points | Yes            | Low                    | Normal      | No             | ← $D$ |
|             | No             | Medium                 | Normal      | No             |       |
|             | No             | Low                    | Abnormal    | Yes            |       |
|             | Yes            | Medium                 | Normal      | Yes            |       |
|             | Yes            | High                   | Abnormal    | Yes            |       |

define: instances, attributes, ground truth

Our first ML task

- Learning to diagnose heart disease ( $T$ )
  - as a (supervised) binary classification task ( $E$ )

our goal: a *classifier*  $h$  : features  $\rightarrow$  labels

data points

| Family History | Resting Blood Pressure | Cholesterol | Heart Disease? |
|----------------|------------------------|-------------|----------------|
| Yes            | Low                    | Normal      | No             |
| No             | Medium                 | Normal      | No             |
| No             | Low                    | Abnormal    | Yes            |
| Yes            | Medium                 | Normal      | Yes            |
| Yes            | High                   | Abnormal    | Yes            |

$\leftarrow D$

e.g. want  $h(\text{No, Medium, Normal}) = \text{No}$



Our first ML task

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our goal: a *classifier*  $h$  : features  $\rightarrow$  labels

data points

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| Yes            | Medium                 | Normal      | Yes            |
| Yes            | High                   | Abnormal    | Yes            |

$\leftarrow D$

Our first ML task

- Learning to diagnose heart disease ( $T$ )
  - as a (supervised) classification task ( $E$ )

|             |                |                        |             |                |
|-------------|----------------|------------------------|-------------|----------------|
|             | features       |                        |             | labels         |
|             | Family History | Resting Blood Pressure | Cholesterol | Heart Disease? |
| data points | Yes            | Low                    | Normal      | Low Risk       |
|             | No             | Medium                 | Normal      | Low Risk       |
|             | No             | Low                    | Abnormal    | Medium Risk    |
|             | Yes            | Medium                 | Normal      | High Risk      |
|             | Yes            | High                   | Abnormal    | High Risk      |

←  $D$



Our first ML task

- Learning to diagnose heart disease ( $T$ )
  - as a (supervised) regression task ( $E$ )

|             |                |                        |             |                                  |
|-------------|----------------|------------------------|-------------|----------------------------------|
|             | features       |                        |             | labels                           |
|             | Family History | Resting Blood Pressure | Cholesterol | Medical cost from heart disease? |
| data points | Yes            | Low                    | Normal      | \$0k                             |
|             | No             | Medium                 | Normal      | \$0k                             |
|             | No             | Low                    | Abnormal    | \$10k                            |
|             | Yes            | Medium                 | Normal      | \$17k                            |
|             | Yes            | High                   | Abnormal    | \$23k                            |

←  $D$

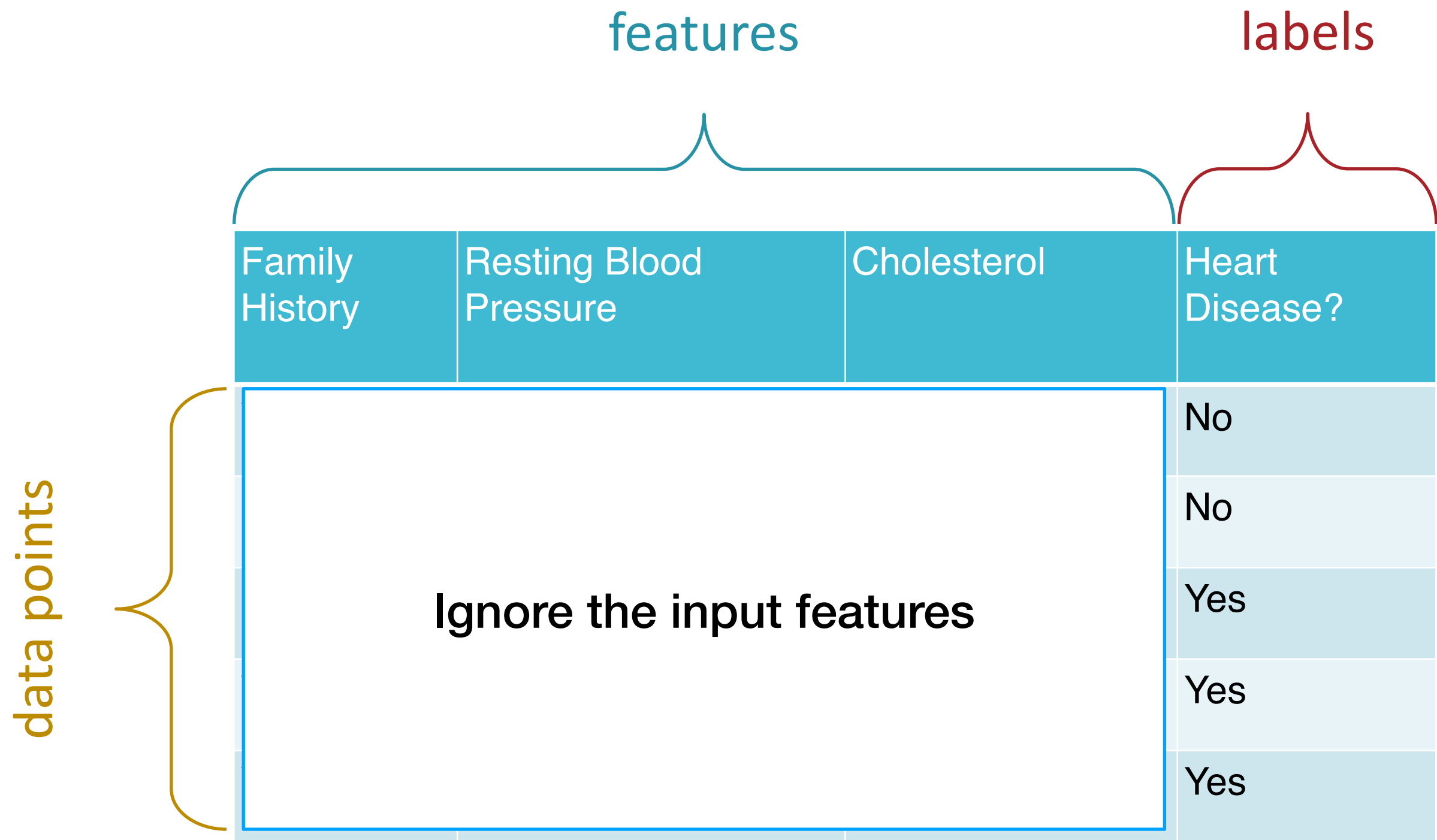
*Our first ML method*

- Majority vote classifier
  - ▶ for a **supervised binary classification task**

|             |                |                        |             |                |
|-------------|----------------|------------------------|-------------|----------------|
|             | features       |                        |             | labels         |
|             | Family History | Resting Blood Pressure | Cholesterol | Heart Disease? |
| data points | Yes            | Low                    | Normal      | No             |
|             | No             | Medium                 | Normal      | No             |
|             | No             | Low                    | Abnormal    | Yes            |
|             | Yes            | Medium                 | Normal      | Yes            |
|             | Yes            | High                   | Abnormal    | Yes            |

*Our first ML method*

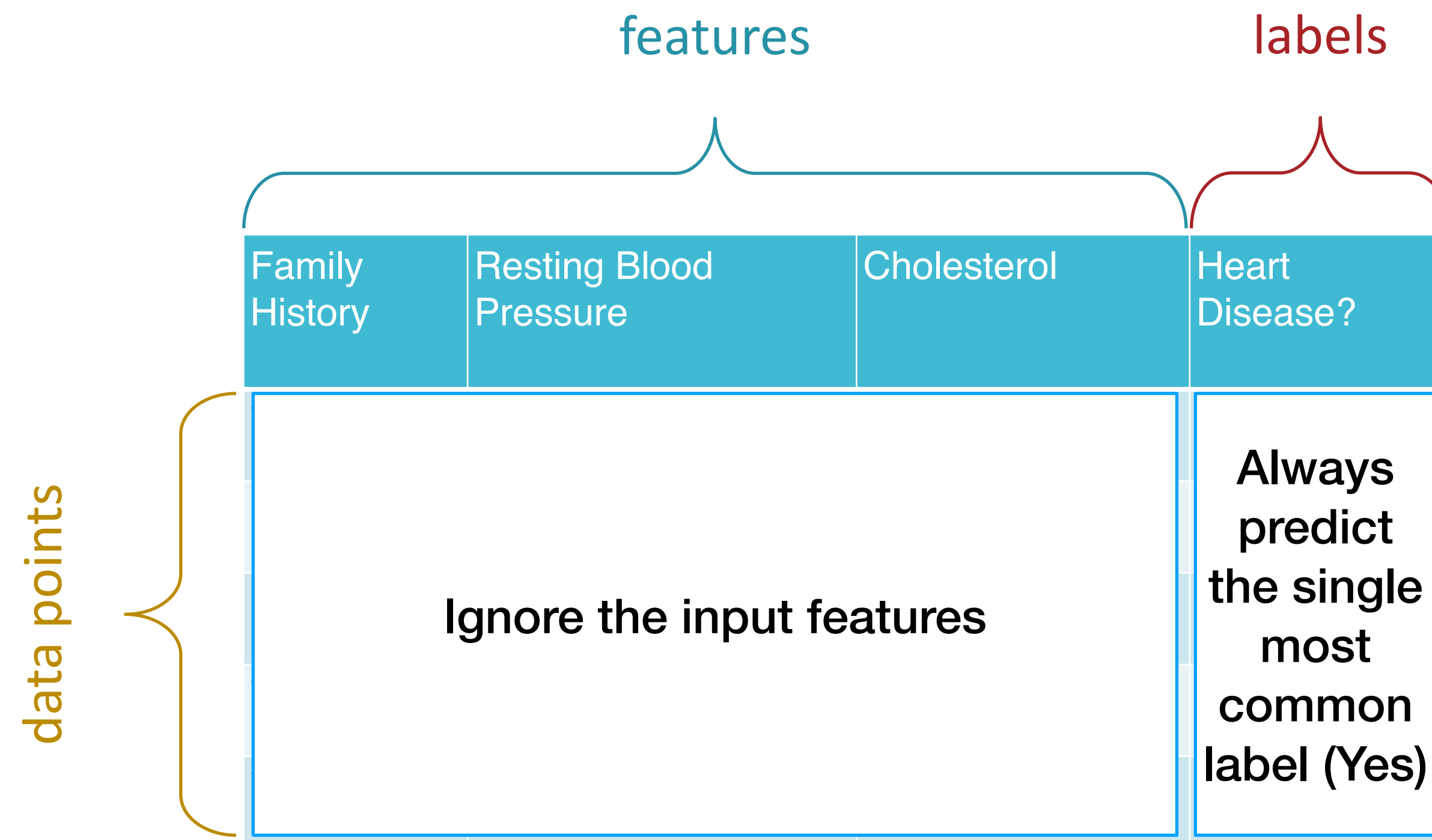
- Majority vote classifier
  - ▶ for a **supervised binary classification task**





# *Our first ML method*

- Majority vote classifier
  - ▶ for a **supervised binary classification task**





*Our first ML method*

- Majority vote classifier
  - for a supervised binary classification task

|             |                |                        |             |                |              |
|-------------|----------------|------------------------|-------------|----------------|--------------|
|             | features       |                        |             | labels         | predictions  |
|             | Family History | Resting Blood Pressure | Cholesterol | Heart Disease? | Model output |
| data points | Yes            | Low                    | Normal      | No             | Yes          |
|             | No             | Medium                 | Normal      | No             | Yes          |
|             | No             | Low                    | Abnormal    | Yes            | Yes          |
|             | Yes            | Medium                 | Normal      | Yes            | Yes          |
|             | Yes            | High                   | Abnormal    | Yes            | Yes          |

error rate =  $\frac{2}{5} = 40\%$  (P)

***Is this  
good?***

- Is 40% error high or low?
- Do we care about this error rate?

No

*Training set  
vs. testing  
set*

|                      |                |                        |             | features       | labels |
|----------------------|----------------|------------------------|-------------|----------------|--------|
| training data points | Family History | Resting Blood Pressure | Cholesterol | Heart Disease? |        |
|                      | Yes            | Low                    | Normal      | No             |        |
|                      | No             | Medium                 | Normal      | No             |        |
|                      | No             | Low                    | Abnormal    | Yes            |        |
|                      | Yes            | Medium                 | Normal      | Yes            |        |
|                      | Yes            | High                   | Abnormal    | Yes            |        |
| testing data points  | No             | Low                    | Normal      | No             |        |
|                      | No             | Medium                 | Normal      | No             |        |
|                      | Yes            | Medium                 | Abnormal    | Yes            |        |

training error rate =  
testing error rate =

$(\hat{P})$   
 $(P)$



# Training set vs. testing set

Majority vote classifier: Always predict the single most common label *in the training set* (Yes)

|                      |     |        |          | features       | labels                 |             |                |
|----------------------|-----|--------|----------|----------------|------------------------|-------------|----------------|
|                      |     |        |          | Family History | Resting Blood Pressure | Cholesterol | Heart Disease? |
| training data points | Yes | Low    | Normal   | No             |                        |             |                |
|                      | No  | Medium | Normal   | No             |                        |             |                |
|                      | No  | Low    | Abnormal | Yes            |                        |             |                |
|                      | Yes | Medium | Normal   | Yes            |                        |             |                |
|                      | Yes | High   | Abnormal | Yes            |                        |             |                |
| testing data points  | No  | Low    | Normal   | No             |                        |             |                |
|                      | No  | Medium | Normal   | No             |                        |             |                |
|                      | Yes | Medium | Abnormal | Yes            |                        |             |                |

training error rate =

testing error rate =

$(\hat{P})$

$(P)$



# Training set vs. testing set

Majority vote classifier: Always predict the single most common label *in the training set* (Yes)

|                      |     |        |          |     | features       | labels                 | predictions |                |              |
|----------------------|-----|--------|----------|-----|----------------|------------------------|-------------|----------------|--------------|
|                      |     |        |          |     | Family History | Resting Blood Pressure | Cholesterol | Heart Disease? | Model output |
| training data points | Yes | Low    | Normal   | No  | Yes            |                        |             |                |              |
|                      | No  | Medium | Normal   | No  | Yes            |                        |             |                |              |
|                      | No  | Low    | Abnormal | Yes | Yes            |                        |             |                |              |
|                      | Yes | Medium | Normal   | Yes | Yes            |                        |             |                |              |
|                      | Yes | High   | Abnormal | Yes | Yes            |                        |             |                |              |
| testing data points  | No  | Low    | Normal   | No  | Yes            |                        |             |                |              |
|                      | No  | Medium | Normal   | No  | Yes            |                        |             |                |              |
|                      | Yes | Medium | Abnormal | Yes | Yes            |                        |             |                |              |

training error rate = 40%  $(\hat{P})$   
testing error rate =  $2/3 \approx 67\%$   $(P)$

Memorizer: if we've seen *exact same* inputs before, predict corresponding output, else majority

*Our second ML method*

| features       |                        |             | labels         |
|----------------|------------------------|-------------|----------------|
| Family History | Resting Blood Pressure | Cholesterol | Heart Disease? |
| Yes            | Low                    | Normal      | No             |
| No             | Medium                 | Normal      | No             |
| No             | Low                    | Abnormal    | Yes            |
| Yes            | Medium                 | Normal      | Yes            |
| Yes            | High                   | Abnormal    | Yes            |

|     |        |          |     |
|-----|--------|----------|-----|
| No  | Low    | Normal   | No  |
| No  | Medium | Normal   | No  |
| Yes | Medium | Abnormal | Yes |

training error rate =  
testing error rate =

Memorizer: if we've seen *exact same* inputs before, predict corresponding output, else majority

*Our second ML method*

|                      |  | features       |                        |             | labels         | predictions  |
|----------------------|--|----------------|------------------------|-------------|----------------|--------------|
|                      |  | Family History | Resting Blood Pressure | Cholesterol | Heart Disease? | Model output |
| training data points |  | Yes            | Low                    | Normal      | No             | No           |
|                      |  | No             | Medium                 | Normal      | No             | No           |
|                      |  | No             | Low                    | Abnormal    | Yes            | Yes          |
|                      |  | Yes            | Medium                 | Normal      | Yes            | Yes          |
|                      |  | Yes            | High                   | Abnormal    | Yes            | Yes          |
| testing data points  |  | No             | Low                    | Normal      | No             |              |
|                      |  | No             | Medium                 | Normal      | No             |              |
|                      |  | Yes            | Medium                 | Abnormal    | Yes            |              |

training error rate = 0%  
testing error rate =



Memorizer: if we've seen *exact same* inputs before, predict corresponding output, else majority

*Our second ML method*

| features       |                        |             | labels         | predictions  |
|----------------|------------------------|-------------|----------------|--------------|
| Family History | Resting Blood Pressure | Cholesterol | Heart Disease? | Model output |
| Yes            | Low                    | Normal      | No             | No           |
| No             | Medium                 | Normal      | No             | No           |
| No             | Low                    | Abnormal    | Yes            | Yes          |
| Yes            | Medium                 | Normal      | Yes            | Yes          |
| Yes            | High                   | Abnormal    | Yes            | Yes          |
| No             | Low                    | Normal      | No             | Yes          |
| No             | Medium                 | Normal      | No             | No           |
| Yes            | Medium                 | Abnormal    | Yes            | Yes          |

training error rate = 0/10  
testing error rate = 1/3      3/3 = 100%



# ***Two different learners***

- Same goal, two different learners
- From the same data, each learner chooses a ***different*** hypothesis
  - ▶ *majority* favors a simpler hypothesis, has similar training and test error
  - ▶ *memorizer* favors a more complex hypothesis, training error is much lower than test
- Difference is called ***inductive bias***
  - ▶ criteria that cause an ML method to favor one hypothesis over another when data isn't enough to determine
  - ▶ “bias” sounds bad, but it is ***unavoidable*** — learning means filling in information that isn't in the data, so we have to have a way to choose

# ***Notation***

- Data point =  $\langle x^{(t)}, y^{(t)} \rangle$
- Feature vector and feature space  $x^{(t)} \in \mathcal{X}$  (e.g.,  $\mathbb{R}^d$ )
  - ▶ individual feature:  $x_j^{(t)}$  (e.g., blood pressure  $\in \mathbb{R}$ , family history  $\in \{0, 1\}, \dots$ )
- Label and label space  $y^{(t)} \in \mathcal{Y}$  (e.g., low or high risk)
- Hypothesis and hypothesis space  $h \in \mathcal{H}$ 
  - ▶ when we say “model” or “classifier” we sometimes mean  $h$  (a specific prediction rule) and sometimes  $\mathcal{H}$  (the space of rules that we’re going to pick from)
  - ▶ hopefully clear from context

# Notation

- **Unknown** target or true classifier  $h^*$ 
  - ▶ best possible  $P$  (test error), might not even be in  $\mathcal{H}$
- Apparent best classifier  $\hat{h} \in \mathcal{H}$  given training set (sometimes called ERM, empirical risk minimizer)
  - ▶ optimizes  $\hat{P}$  (here training error) over  $\mathcal{H}$
- ML method: a way of picking a hypothesis given data
  - ▶ e.g., majority vote classifier: empirical risk minimization over constant classifiers



A classifier that's used in practice: mini mental state exam (MMSE)

1. Time: 10 seconds for each reply:

- a) What year is this? (accept exact answer only).
- b) What season is this? (accept either: last week of the old season or first week of a new season).
- c) What month is this? (accept either: the first day of a new month or the last day of the previous month).
- d) What is today's date? (accept previous or next date).
- e) What day of the week is this? (accept exact answer only).

features

|   |    |
|---|----|
| 1 | /1 |
| 1 | /1 |
| 0 | /1 |
| 0 | /1 |
| 1 | /1 |

... 30 total questions like these

score = number of points =  $\sum_{j=1}^{30} x_j^{(t)}$

decision rule

Mini-Mental State Exam Scoring Chart

| Score         | Level of Dementia             |
|---------------|-------------------------------|
| 24 and higher | Normal cognition; no dementia |
| 19 – 23       | Mild dementia                 |
| 10 – 18       | Moderate dementia             |
| 9 and lower   | Severe dementia               |

threshold



# ***MMSE is a linear classifier***

- Linear classifier (aka linear separator, linear discriminant, linear decision rule, halfspace)
- Compute a score (aka discriminant function) by assigning each feature a number of points (aka weight)

$$\text{score of example } t = \sum_{j=1}^d w_j x_j^{(t)}$$

*MMSE has  $w_j = 1 \ \forall j$*

- Compare score to a threshold:
  - ▶ e.g., normal cognition iff score  $\geq 24$
- Called a linear classifier because *score* is linear in *weights*
  - ▶ the decision is *not* linear in anything (threshold)
  - ▶ later we'll allow nonlinear transformation of features — still a linear classifier since it's linear in weights
- Nonlinear classifier: score (discriminant) is a general fn  $f_w(x^{(t)})$  or  $f(x^{(t)}; w)$  — still has learned parameters  $w$

*Other classifiers can have negative or fractional weights*

# ***Learning goals***

- You should be able to
  - ▶ Formulate a well-posed learning problem for a real-world task by identifying the task, performance measure, experience, and data
  - ▶ Describe the supervised learning paradigm in terms of the type of data needed, the form of prediction, and the structure of the output prediction
  - ▶ Explain the difference between memorization and generalization (relatedly, overfitting and underfitting)
  - ▶ Explain the difference between training and testing data